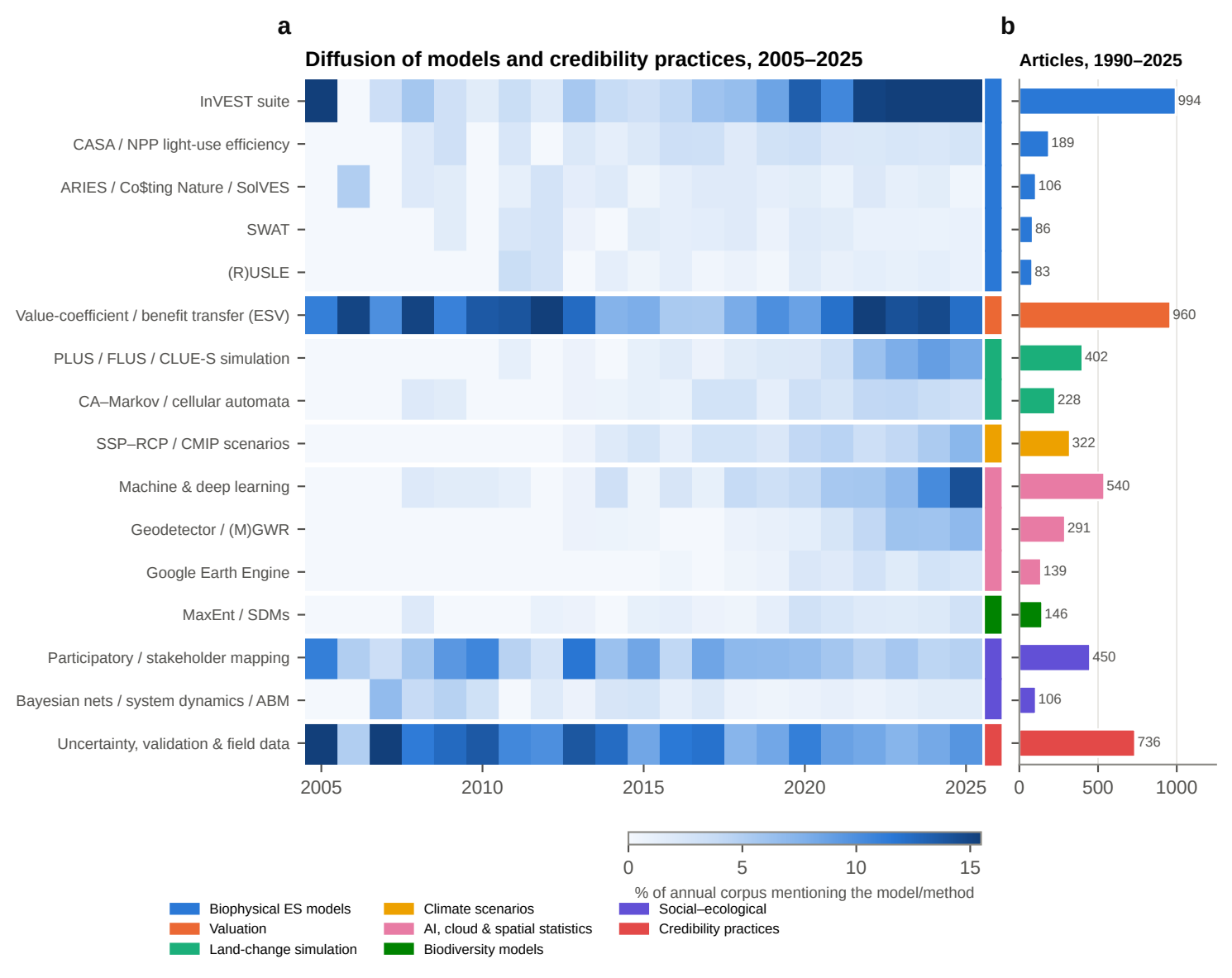

a
Diffusion of models and credibility practices, 2005–2025
b
Articles, 1990–2025
InVEST suite
CASA / NPP light-use efficiency
ARIES / Co$ting Nature / SolVES
SWAT
(R)USLE
Value-coefficient / benefit transfer (ESV)
PLUS / FLUS / CLUE-S simulation
CA–Markov / cellular automata
SSP–RCP / CMIP scenarios
Machine & deep learning
Geodetector / (M)GWR
Google Earth Engine
MaxEnt / SDMs
Participatory / stakeholder mapping
Bayesian nets / system dynamics / ABM
Uncertainty, validation & field data
994
189
106
86
83
960
402
228
322
540
291
139
146
450
106
736
2005
2010
2015
2020
2025
0
500
1000
0
5
10
15
% of annual corpus mentioning the model/method
Biophysical ES models
Valuation
Land-change simulation
Climate scenarios
AI, cloud & spatial statistics
Biodiversity models
Social–ecological
Credibility practices